

9. Which group are called the 'amphibians' of the plant kingdom?

- (a) Gymnosperms (b) Pteridophytes
(c) Bryophytes (d) Algae

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Ans. (c)

Bryophytes are called the amphibians of the plant kingdom. Amphibians are those organisms which live on both land and in water.

10. Which of the following is an artificial ripening agent?

- (a) Ethylene (b) Ethephon
(c) 2, 4-D (d) NAA

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Ans. (a) & (b)

Ethephon and ethylene both are useful as artificial ripening agents, the question is deleted.

III. Plant Reproduction

1. Embryo is found in –

- (a) Flowers (b) Leaves
(c) Seeds (d) Buds

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The embryo along with other cells from the mother plant develops into the seed for the next generation, which, after germination, grows into a new plant.

2. Plants are formed like parent plant from –

- (a) Seeds (b) Stem cut
(c) None of these (d) Both of these

39th B.P.S.C. (Pre) 1994

Ans. (b)

Plants are propagated by two methods –

1. Sexual propagation or seed propagation.
2. Asexual or vegetative propagation.

Vegetative propagation is a form of asexual reproduction of a plant. Only one plant is involved and offspring is the result of one parent. The new plant is genetically identical to the parent. In this process, new organism arises from any vegetative part of the parent (root, stem, leaf etc.) and possesses exactly the same characteristics of the parent plant. The purity of plants is maintained in vegetative propagation which is not possible in sexual reproduction. Stem cutting, grafting, layering, etc. are common artificial methods of vegetative propagation.

3. Stem cutting is done generally to grow –

- (a) Banana (b) Sugarcane
(c) Mango (d) Cotton

39th B.P.S.C. (Pre) 1994

Ans. (b)

Botanically, Sugarcane belongs to the Andropogonae tribe of the family Gramineae. Sugarcane plants are propagated by stem cutting. Stem cutting is segmented parts of the stem or entire stems detached from the main branch. Mango and cotton are generally grown from seeds while suckers are used as the planting material for banana. Sucker in banana arises from the root and is an extension of the rhizome as a lateral root. Some farmers also practice tissue culture for banana cultivation using seedlings developed through tissue culture. Rhizomes and peepers are the other planting materials used for banana cultivation.

4. The stem of the plant helps in distributing food to all parts of the plant. It also helps in –

- (a) Storing of food (b) Shaping the plant
(c) Respiration (d) Photosynthesis
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (e)

The stem of the plant helps to distribute food (nutrients along with water and store food (nutrients) to all parts of the plant. It plays the main role to supports the leaves, flowers and fruits of the plant as well as gives shape to the plant.

5. Pollination is best defined as –

- (a) transfer of pollen grain from anther to stigma
(b) germination of pollen grains
(c) growth of pollen tube in ovule
(d) visiting flowers by insects
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

In pollination, the pollen grains of the flowers's anther are transferred to the pistil or stigma of the flower. This process is called pollination.

6. The anther contains

- (a) pollen grains (b) sepals
(c) ovules (d) carpels

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Ans. (a)

The anther usually contains four pollen sacs which are responsible for producing pollen grains. Each pollen grain is a single cell containing two male.